TÜBİTAK 123E294 – No-flight simulation results (Scenario I & II)

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Summary

With paper-like noise, the corrected sphere method (Scenario II) localises a ground target 40-100 m away to a median of 4.4 m with R = 5 m and 1.0 m with R = 20 m. The repo version as found gives about 70 m. With noise measured in our own field logs, Scenario II at these distances stays at 60-75 m. Scenario I ("+" formation, direct tracking) still reaches a median of 6.4 m (quiet field noise) and 10.6 m (typical field noise).

- **Scenario II is limited by geometry and multipath, not by the filter.** The largest possible P3-P1 difference is only 0.76 dB at R = 5 m, d = 70 m. Field noise is 1.2-4 dB per sample, and spatial multipath biases the bearing whatever filter is used.
- **The fixes matter more than the filter.** Geometry fix, error normalisation, window convergence, bias-cancelling flip and a lateral baseline take the paper-like error from 71 m to 4.4 m (R = 5 m).
- **Filter choice changes little.** Raw, Kalman, SISMF-signal and moving average give the same Scenario II accuracy within Monte Carlo spread. In Scenario I, Kalman + P (K = 0.5) gives the same accuracy as the SISMF axis controller with less than half the flown path.
- **Safety: the paper's x-z vertical circle can put two drones on the same point.** Two commanded slots came closer than 3 m in 20-100 % of paper-geometry runs, depending on settings. The yaw-aligned vertical circle keeps them at least R·√2 apart.
- **For the report:** state Scenario II's accuracy together with its noise level and R/d, and present Scenario I as the method that works at field noise today.

What the repo and the papers contain

The repo `enesmstf/tubitak_drone` holds the current code, nine field logs, three calibrations and the hoca's documents in `guncel/formuller/`. Five formula sources were used:

Source	What it gives	Reported result
Progress report 2 (Scenario I)	Scalar Kalman (Eq. 4-8), "+" formation driven to zero difference	11.4 m raw → 2.3 m with Kalman
Progress report 2 (Scenario II)	ISMF angle law (Eq. 12-17), PID angle law (Eq. 32), augmented point (Eq. 36)	Kalman cuts error by 61 %
IEEE Access 2024 (ISMF)	$s = \dot{e} + \lambda e$, $\dot{\alpha} = k \cdot \mathrm{sign}(s) + \int e$, line $c + t \cdot (v1 \times v2)$ (Eq. 16-17)	23.7 m mean, 100 Monte Carlo, 4 % noise
IJSS 2026 (SISMF, SCI_4)	$s = e + \lambda \int e$, $\dot{\alpha} = k \cdot \mathrm{sign}(s) + \kappa \int e$, $\lambda = 1$, $k = 0.2$, $\kappa = 0.05$; least-squares ray fusion (Eq. 8)	12-15 m (ground rods, R = 50 m)
Progress report 3	LoRa + SDR integration, no new formulas	—

Four corrections came out of reading them against each other:

1. **D2/D4 typo.** Every document prints $D2 = c - R(\sin\beta, 0, \cos\beta)$ and $D4 = c + R(\cos\beta, 0, \sin\beta)$. These two points are not $2R$ apart and do not match $v2 = R(\cos\beta, 0, \sin\beta)$. The consistent form is $D2/D4 = c \mp R(\cos\beta, 0, \sin\beta)$. `ground_station_core_sphere.py` still uses the typo.
2. **The augmented point is not needed.** Equal power on a pair puts the target on that pair's perpendicular-bisector plane. Both planes pass through c, so their intersection is exactly the line $c + t \cdot (v1 \times v2)$. The augmented point lies on the same line; the earlier two-branch search adds nothing.
3. **Eq. 17 drops the centre.** It should read $x = x_c + R^2 \sin\alpha \sin\beta\, t$ (likewise y, z); the printed form omits x_c and y_c.
4. **The 2026 gains assume an angle-unit error.** In the IJSS simulation e is an angle error. Fed with a dB difference, $\lambda = 1$ and $\kappa = 0.05$ drive a ±100° limit cycle in our simulator.

Two code findings beyond the formulas:

- `ground_station_core_kalman.py` uses old offsets {−2.1, +1.9, −0.1, +0.2} dB, not the accepted field calibration {0, −1.32, +2.38, +1.10} dB.
- `ground_station_core_sphere.py` records each line at the centre *after* it has been shifted. It also shifts 3 m *along* the line, so successive lines are nearly parallel and their intersection is ill-conditioned.

Simulation model

The simulator (`sphere_sim/`, pure numpy) replaces the radio, the drones and LoRa with models taken from our own calibration and logs; everything else follows the papers and the repo code.

$$P_i = P_{ref} - 10\,n\,\log_{10} d_i + S(\mathbf{x}_i) + b_i + w_i(k), \qquad w_i(k) = \rho\, w_i(k-1) + \sqrt{1-\rho^2}\,\sigma_t\,\varepsilon$$

- P_ref = −78.72 dBm and n = 1.264 from `kalibrasyon_sonucu_saha.json`.
- S(x) is spatial shadowing (multipath), a Gaussian random field with std σs and correlation length Lc. It is the same for all drones and fixed in space.
- b_i is the residual calibration error per drone after the offsets are applied (std 0.5 dB).
- w_i is temporal noise, AR(1). The field logs give 1.2-4 dB around a 30 s median and lag-1 correlation 0.3-0.6.
- Raw values are rounded to 1 dB before the offsets are added, one sample per drone every 2 s, as in the logs.
- Drones fly to commanded slots at 3 m/s with 0.5 m position noise. Control runs every 4 s (Scenario II) or 5 s (Scenario I) on the window mean of filtered samples, as `get_windowed_rssi` does.

Noise scenario	σt (dB)	ρ	σs (dB)	Lc (m)	Basis
paper-like	0.17	0	0	—	IEEE Access 2024: "4 % variance in signal power" read as ≈0.17 dB
field-temporal	2.5	0.5	0	—	typical logs, multipath switched off to isolate the filters
field-quiet	1.2	0.3	1.0	20	quietest log (`1791809751`)
field-typical	2.5	0.5	1.5	20	logs `1792067603`, `1792069619`

The shadowing values σs and Lc are assumptions. Our logs were taken on the ground with a hand-held radio, so they cannot separate airborne multipath from temporal noise.

Targets lie on the ground at 40-100 m from the first sphere centre (Scenario II) or 20-40 m from the start (Scenario I), at random bearings. Each configuration uses 40-60 Monte Carlo runs with fixed seeds.

Scenario II (sphere) results

Scenario II works under paper-like noise once the code is corrected, and fails at 40-100 m under field noise for every filter and angle law tested.

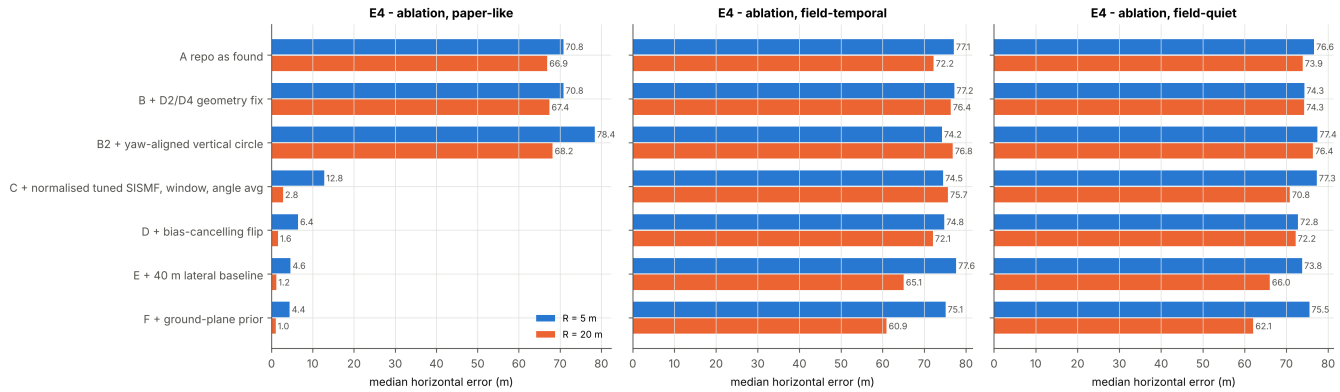
Paper reproduction (E1)

Paper-like noise, R = 5 m, the three fixed centres (0,0,15), (30,10,15), (0,30,15), 60 runs per row.

Angle law	Repo gains, |e| < 1 dB (median 3-D error)	Repo gains, |e| < 0.02 dB	Tuned, window criterion	Line error, tuned
PID (Report 2)	74.6 m	78.8 m, never converged	16.0 m	4.1°
ISMF-2024	68.6 m	75.7 m, never converged	11.1 m	3.2°
SISMF-2026	68.5 m	80.5 m, never converged	10.1 m	3.4°

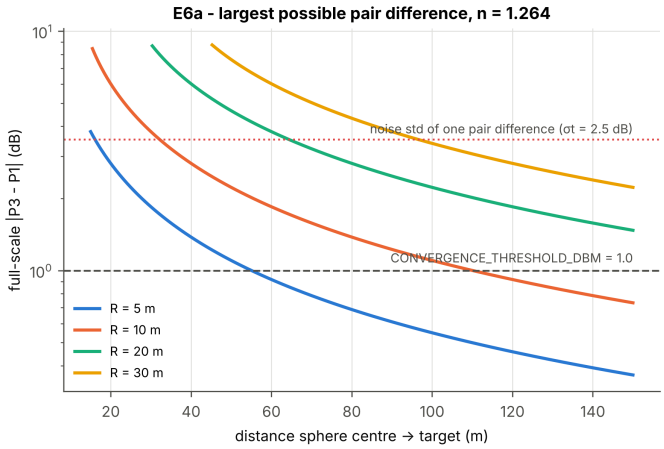
With the repo's 1 dB threshold every centre "converges" within about 20 s, before the angles have moved. At 70 m with R = 5 m the largest possible |P3 − P1| is 0.76 dB, below the 1 dB threshold (figure E6 below). The tuned laws reach 10-16 m, and the three laws differ by less than the Monte Carlo spread once tuned the same way. The paper reports 23.7 m mean; our tuned SISMF gives 21.6 m mean, consistent with it.

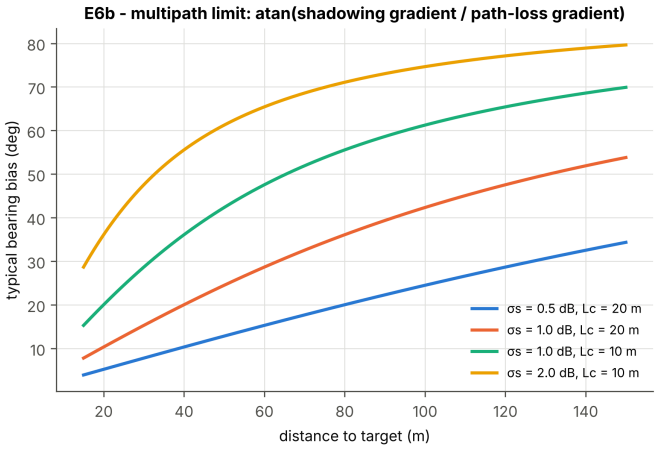
What each fix contributes (E4)



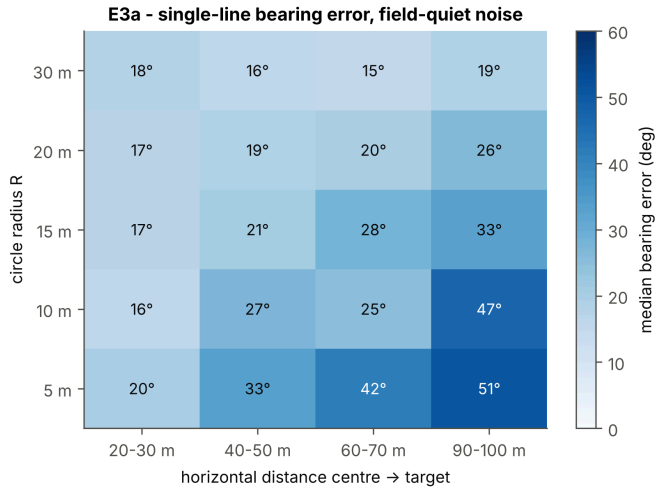
With paper-like noise the error falls from 71 m to 4.4 m (R = 5 m) and from 67 m to 1.0 m (R = 20 m). Step C (normalised error, window criterion, angle averaging) does most of the work. The flip halves it again by cancelling per-drone offset errors, and the lateral baseline removes the near-parallel lines. With field noise no step brings the error below 60 m; the per-line error does fall, from 48° to 32° at R = 20 m.

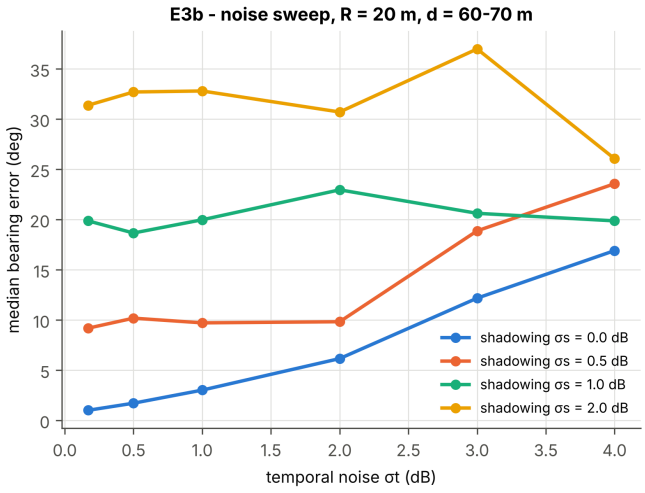
Why field noise defeats it (E3, E6)





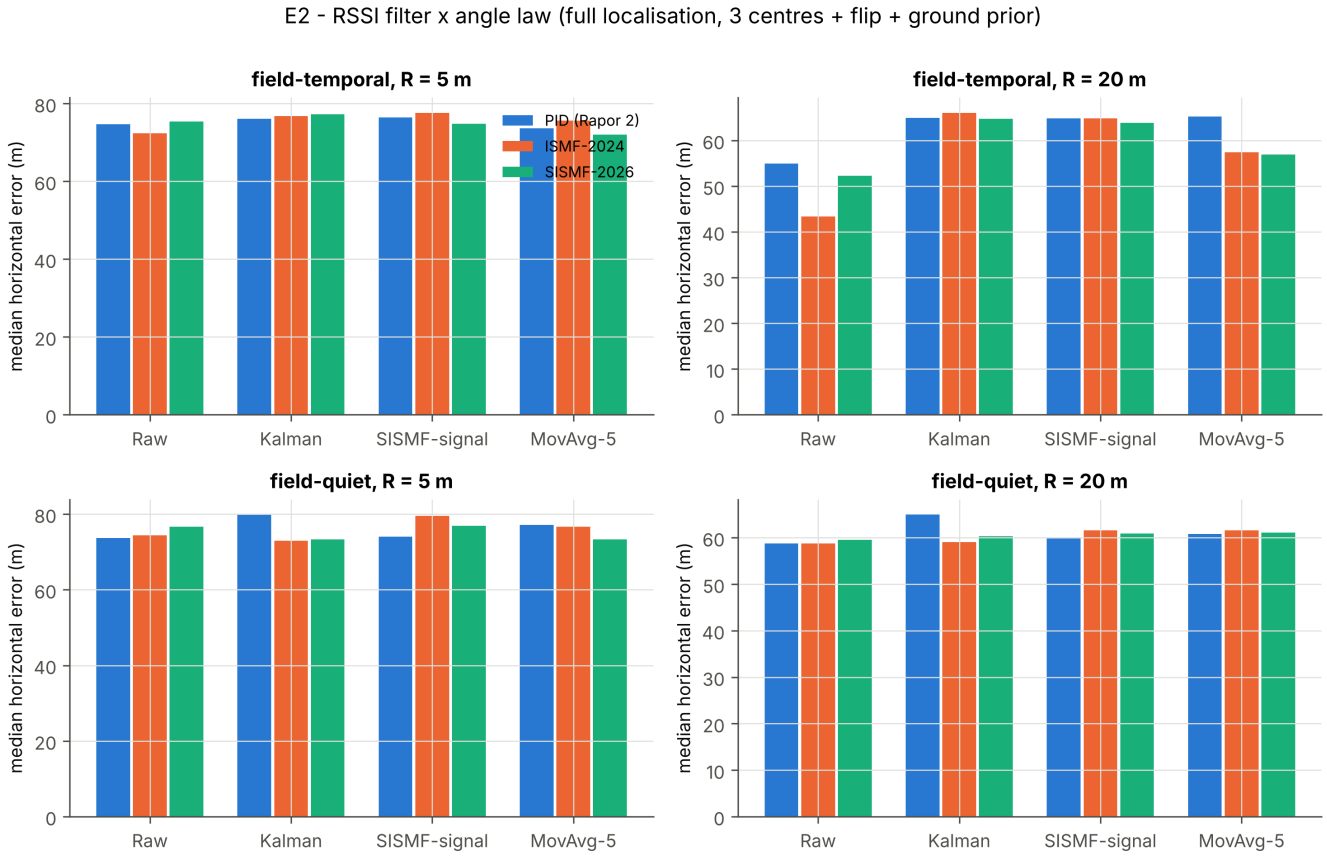
The usable signal is the pair difference, about 8.7·n·R/d dB, which falls below one dB beyond 55 m at R = 5 m. A spatial shadowing gradient (σs/Lc dB/m) tilts the equal-power direction by atan of its ratio to the path-loss gradient, 4.34·n/d dB/m. That bias is independent of R and of any time filter.





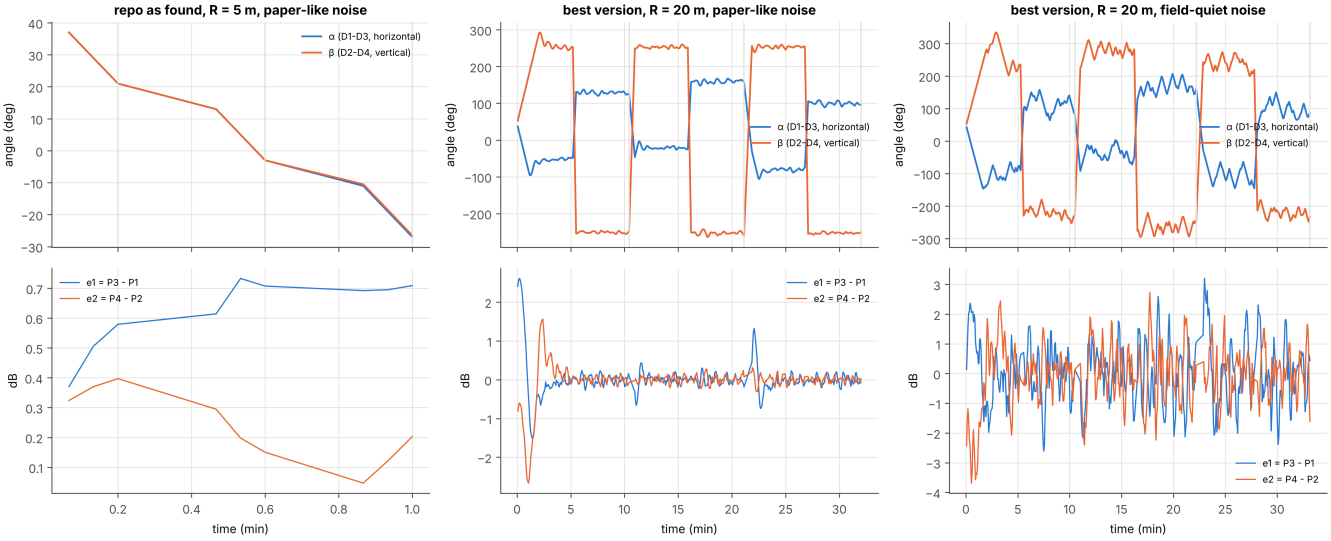
Without multipath (σs = 0) the bearing error stays at 1-6° up to σt = 2 dB at R = 20 m, d = 60-70 m. Adding only 0.5 dB of shadowing raises it to about 8-10°, and 1 dB to 21-28°. Larger R and closer targets help: R = 30 m keeps the bearing error at 14-19° from 20 to 100 m, against 20-51° at R = 5 m.

Filters and angle laws (E2)

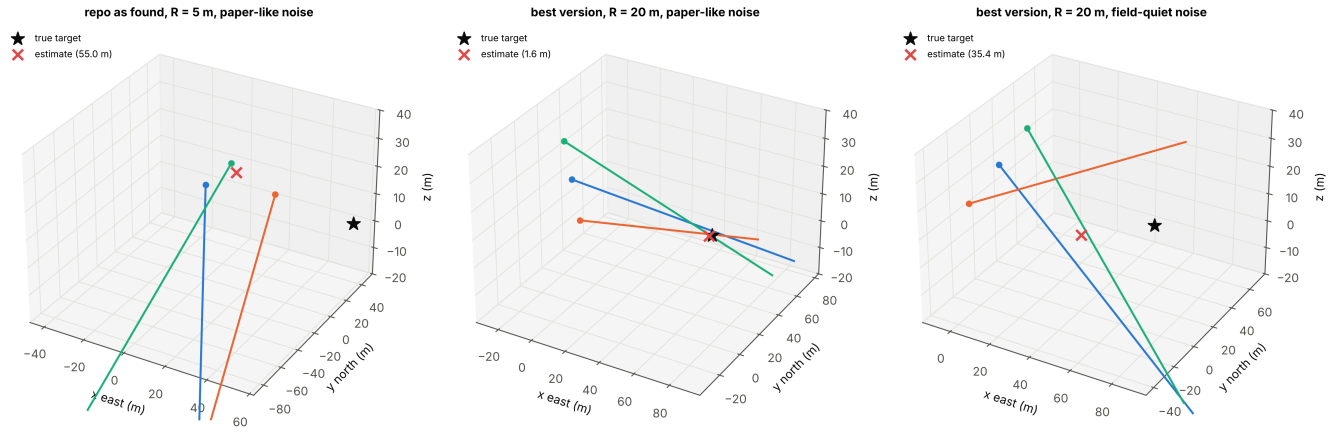


No RSSI filter or angle law stands out. At R = 5 m all twelve combinations sit at 72-80 m. At R = 20 m (field-temporal) raw RSSI does slightly better (43-55 m) than Kalman or SISMF-signal (64-66 m). The extra filter lag inside the angle loop appears to cost more than its smoothing gains, since the angle law already integrates.

Example runs (E7)



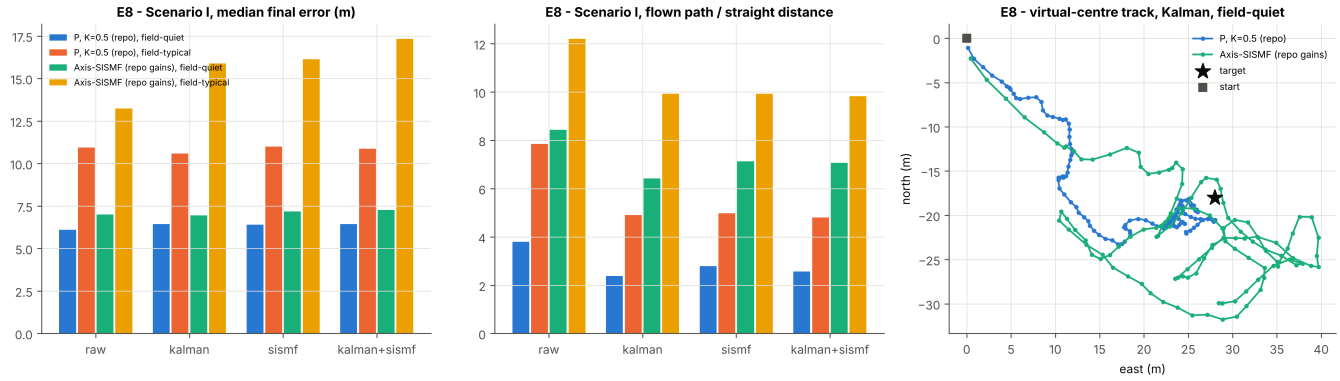
The repo version (left) stops after about one minute with e still near 0.7 dB. The best version under paper-like noise settles α and β at each centre; the jumps of 180° are the deliberate pair flips. Under field-quiet noise (right) the angles wander by ±50° around the equilibrium.



Scenario I ("+" formation) results

Kalman + the proportional step (K = 0.5 m/dB, the Report 2 law) is the best Scenario I setting: same accuracy as the SISMF axis controller, about half the flown path and half the hover jitter. Results below are medians of 40 runs, target 20-40 m from the start, 10 min of tracking.

Noise	Filter + controller	Final error (m)	Reached < 5 m	Time to 5 m (s)	Hover jitter (m)	Path / straight line
field-quiet	Kalman + P (repo)	6.4	73 %	180	1.9	2.4
field-quiet	Kalman + axis-SISMF (repo)	6.9	85 %	86	4.1	6.4
field-quiet	Raw + P	6.1	75 %	180	1.9	3.8
field-typical	Kalman + P (repo)	10.6	73 %	186	4.6	4.9
field-typical	Kalman + axis-SISMF (repo)	15.9	80 %	128	8.3	9.9
field-typical	Kalman + SISMF-signal + axis-SISMF (v1 default)	17.3	75 %	140	8.9	9.8



- The RSSI filter changes the final error by under 1 m. Its benefit is a smoother track: Kalman cuts the flown path from 3.8× to 2.4× the straight line (field-quiet, P).
- The axis-SISMF controller reaches the target about twice as fast, but its integral term keeps the swarm circling 4-8 m around it. That costs battery and widens the hover footprint.
- 15-27 % of runs never get within 5 m with either controller; the 90th-percentile final error is 25-45 m. Near the target, the 15 m altitude flattens the horizontal gradient, and multipath creates false equilibria. This is the same limit as Scenario II, but at shorter range, where the path-loss gradient is stronger.
- With paper-like noise every setting ends within 0.3-2.7 m, matching the 2.3 m of Report 2.

RSSI filter benchmark

Against a known true RSSI, Kalman (Q = 1, R = 20) is the most accurate filter whenever the radio moves. SISMF used as an RSSI smoother matches it on a steady signal but lags longer: 1.3 times on a ramp and 3 times after a step. Lower output variance, the metric in the earlier SISMF-vs-Kalman report, does not show which filter is closer to the truth.

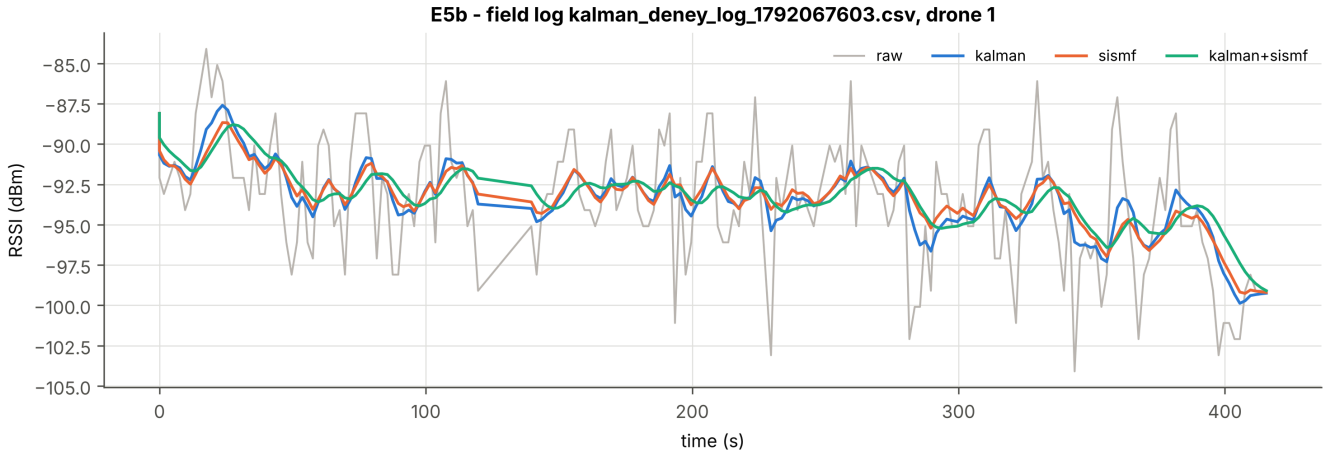


E5 runs 300 Monte Carlo series with field-typical noise (σt = 2.5 dB, ρ = 0.5, integer dB):

Filter	RMSE vs truth, steady (dB)	RMSE vs truth, moving radio (dB)	RMSE vs truth, +12 dB step (dB)	Lag, moving radio (s)	Lag, step (s)	Output-variance cut, moving radio
Raw	2.52	2.52	2.52	0.8	0.0	—
Kalman	1.26	1.74	1.72	7.8	5.6	9 %
SISMF-signal (repo gains)	1.26	1.89	2.40	9.8	16.9	11 %
Kalman + SISMF	1.15	2.28	2.42	13.5	18.0	11 %
Moving average (5)	1.66	1.79	1.93	4.0	4.1	5 %

On the moving radio, SISMF and Kalman + SISMF report the largest variance cut yet sit further from the true RSSI than Kalman. The SISMF smoother is rate-limited at k = 0.3 dB/s, so a 12 dB step takes about 17 s to follow.

On the nine field logs (16 drone series, E5b) the same pattern holds. Mean output variance is 18.4 dB² raw, 6.9 Kalman, 5.0 SISMF and 4.9 Kalman + SISMF. Lag against a centred 30 s median is 7.3 s for Kalman, 9.8 s for SISMF and 12.7 s for the chain. That reference itself reads about 1.9 s for raw data, so the lags carry roughly ±2 s of method bias.



Suggested wording for the report: SISMF smooths the RSSI more than Kalman (variance −73 % vs −63 % on the field logs), at the price of a longer lag. Against a known signal Kalman is the more accurate estimator when the source or the drones move. The chained filter adds lag without adding accuracy.

Best versions of the code

Both v2 files are generated from the repo files by scripts that apply each change as an explicit text replacement, so every edit can be reviewed. Both compile; Scenario II v2 was run end to end in its new simulation mode.

Scenario II – `ground_station_core_sphere_v2.py`

1. D2/D4 = c ∓ R·b (typo fixed).
2. `VERTICAL_PLANE = "aligned"`: the vertical circle is yawed perpendicular to the D1-D3 axis, so drones stay at least R·√2 apart. `"fixed_xz"` keeps the paper geometry.
3. `NORMALIZE_ERROR`: e is divided by the full-scale difference predicted from the calibration, so the loop gain no longer depends on distance. Tuned gains: SISMF λ = 0.01, k = 0.2, κ = 0.0016, φ = 1.
4. Convergence: at least 300 s per centre, then the mean |e| over the last 30 cycles must be below 0.15 dB; timeout 600 s. The line direction is the axial mean over that window.
5. `FLIP_PAIRS`: each pair is swapped and converged again; averaging the two lines cancels per-drone offset errors.
6. Lines are stored at the centre where they were measured. Centres follow a lateral baseline (`CENTER_PLAN = "baseline"`, ±20 m) instead of 3 m steps along the line. The RSSI-trend sign guess is removed.
7. Final estimate: least-squares intersection with a ground-plane option (`GROUND_PRIOR`).
8. `SIMULATION_MODE` now flies virtual drones in a calibrated channel instead of sending random RSSI. `run_sphere_simulation_v2.py` runs the real ground-station code without hardware, time-scaled. Five runs at R = 20 m, σt = 1.2 dB gave 7-26 m (median 14 m), in line with the numpy simulator's 17 m median for the same case.

Scenario I – `ground_station_core_kalman_v2.py`

1. Calibration offsets set to the accepted field values {0, −1.32, +2.38, +1.10} dB.
2. `SIGNAL_FILTER` and `AXIS_CONTROLLER` are separate switches; the default is Kalman + P (K = 0.5). The CSV still logs all four filters.
3. The `[KARAR MOTORU]` log-line precedence bug is fixed.

Setting	v1 (repo)	v2 default
Scenario II vertical plane	fixed x-z (typo positions)	yaw-aligned
Scenario II convergence	|e| < 1 dB for 3 cycles	window mean < 0.15 dB after 300 s

Setting	v1 (repo)	v2 default
Scenario II centres	3 m along the line	±20 m lateral baseline
Scenario II R	5 m	5 m (15-20 m recommended if the field allows)
Scenario I filter / controller	Kalman + SISMF chain / axis-SISMF	Kalman / P
Calibration offsets (Scenario I)	old ad-hoc values	field calibration

Verified, not verified, next steps

The simulation results are verified as simulations: every number above comes from code that was run, with fixed seeds, and the two independent implementations (numpy simulator and v2 ground-station code) agree. None of it has flown.

Item	Status
D2/D4 typo, collision geometry, augmented-point equivalence	Verified (algebra + simulation)
Scenario II accuracy under paper-like noise	Verified in simulation
Field noise level and correlation	Measured from 9 ground logs; airborne values unknown
Spatial multipath σs, Lc	Assumed; dominates Scenario II results
v2 sphere code	Runs end to end in simulation mode; not run on Pixhawk/SITL
v2 Scenario I code	Compiles and runs in mock mode; not flown
Tuned gains	Simulation only; 4 s control period, 2 s sampling

1. **Hover test to measure airborne noise and multipath (most important).** Hold the four drones at 15 m for 5 min at three positions 20-40 m from a fixed radio, then step the formation angle in 15° increments. This gives σt, ρ, σs and the true e(α) curve, and replaces the assumed values above.

2. **SITL run of `ground_station_core_sphere_v2.py`** with `VERTICAL_PLANE = "aligned"` to check slot tracking and the 3 m separation guard before any real flight.
3. **Choose R from E3.** If 2R = 30-40 m is safe at the test site, R = 15-20 m roughly halves the bearing error compared with R = 5 m.
4. **Fly Scenario I first** (Kalman + P), since it is the method that holds up at field noise; use Scenario II for a coarse bearing at longer range.
5. **Flight time.** The best Scenario II run takes 30-50 min of dwell for three centres with flips, beyond one battery. Splitting centres across batteries, or dropping the flip when offsets are well calibrated, are the options.

Code and results: `sphere_sim/` (simulator, `run_experiments.py`, all CSV tables and figures) and `best_version/` (v2 files, launchers, generator scripts), delivered with this document.